MCS Saturday 7th-SHSAT Math Homework

Instructor: Hsiao 'Hue' Weng | weng.hue.8@gmail.com

Grade 7 SHSAT Math · CAT Digital Format

Percent Decision Lab

Session 3 Homework Activity Sheet · All-New Questions

Name _____ **Date** _____

Completion Check	Core	Transfer	Total
Accuracy + denominator evidence + CAT commit routine	_____ /18	_____ /8	_____ /26

A CAT Commit Protocol

5 prompts · 7 min

Practice the forward-only decision routine. Give observable actions rather than general advice.

Read the ask | **Mark the start** | **Predict size** | **Verify form** | **Commit once**

A1 · Commit cue

Recall

You have solved an item and checked the requested quantity. What exact action ends your work on that screen?

Name the action and what happens next.

Select Next after verifying the recorded response. Advancing submits the answer; I cannot return.

A2 · Adaptive meaning

Explain

Your next item feels easier than the previous one. Explain why one item's difficulty does **not** determine your final result.

Refer to ongoing score estimation, not one question.

The estimate uses the response pattern and item difficulties across the test, not how one item feels.

A3 · Typed form

Apply

Your work gives **0.285**, and the screen asks for the numerical value of the percent. What should you enter if the percent sign is not permitted?

Enter one value **28.5**

Enter only the requested numerical value.

A4 · Denominator check

Reason

Before advancing on a percent-change item, what phrase in the story should you connect to the denominator?

State the phrase and why it controls the denominator.

Use "original," "from," or "before the change": percent change compares the change with that starting value.

A5 · Pace decision

Defend

After about two minutes, your setup is complete but your arithmetic is unfinished. Write a two-sentence rule for making the best supported response and moving forward without panic.

Sentence 1: finish or estimate. Sentence 2: verify form and commit.

I will keep my valid setup and use a short calculation, benchmark, or elimination step to reach the best-supported response. I will check the requested quantity and entry form, then select Next calmly instead of restarting or planning to return.

B Benchmark Warm-Up

6 questions · 9 min

Write a 10%, 5%, 1%, or equivalent-fraction benchmark before calculating.

10% = divide by 10 · 5% = half of 10% · 1% = divide by 100 · Predict before computing

B1 Level 1

Find **10%** of **730**.

Show the decimal-place shift

10% = 1/10
730 ÷ 10 = 73

Typed entry 73

B2 Level 1

Find **5%** of **340**.

Start from 10%

10% of 340 = 34
5% is half: 34 ÷ 2 = 17

Typed entry 17

B3 Level 1

Find **1%** of **8,400**.

Name the place-value move

1% = 1/100
8,400 ÷ 100 = 84

Typed entry 84

B4 Level 2

Find **15%** of **260**.

Combine two benchmark parts

15% = 10% + 5%
26 + 13 = 39

Typed entry 39

B5 Level 2

Which is the best estimate of **32%** of **190**?

A. 6 **B.** 19

C. 61 **D.** 158

Use 30% as an anchor

30% of 190 = 57; 2% = 3.8.
32% of 190 = 60.8 ≈ 61.
Answer: C — 61

B6 Level 2

Write **42.5%** as both a decimal and a fraction in lowest terms.

Percent → decimal → fraction

42.5 ÷ 100 = 0.425
425/1000 = 17/40

Decimal; fraction 0.425; 17/40

C Core Accuracy Set

12 questions · 24 min

Treat each card as one forward-only screen: identify the whole or original, predict, solve, scan the response form, and commit.

Amount = percent × whole · Percent = part ÷ whole · Change ÷ original · Magnitude is nonnegative

C1 Level 2

Find **24%** of **350**.

Decompose 24% or multiply

24% = 25% − 1%
87.5 − 3.5 = 84

One number 84

☐ I checked that my answer is less than 350.

C2 Level 3

Find **42.5%** of **160**.

Use 40%, 2.5%, or an equivalent fraction

42.5% = 17/40
160 × 17/40 = 4 × 17 = 68

One number 68

☐ I compared my result with half of 160.

C3 Level 2

96 is what percent of **320**?

Identify part and whole before dividing

96/320 = 3/10 = 0.30
0.30 × 100 = 30%

Percent number only 30

☐ I converted the quotient to a percent.

C4 Level 3

A quantity rises from **175** to **217**. Find the percent increase.

Change ÷ original × 100

217 − 175 = 42
42/175 = 6/25 = 24%

Percent number only 24

☐ I used 175 as the denominator.

C5 Level 3

A reservoir level falls from **560** centimeters to **434** centimeters. Find the percent decrease.

Estimate before exact division

560 − 434 = 126
126/560 = 9/40 = 22.5%

Percent number only 22.5

☐ I reported a positive magnitude.

C6 Level 2

A value increases by **27** from an original value of **180**. What is the percent increase?

The change is already given

Change = 27; original = 180.
27/180 = 3/20 = 15%

Percent number only 15

☐ I did not subtract 27 again.

C Core Accuracy Set (continued)

12 questions · 24 min

Treat each card as one forward-only screen: identify the whole or original, predict, solve, scan the response form, and commit.

Amount = percent × whole · Percent = part ÷ whole · Change ÷ original · Magnitude is nonnegative

C7 · Multi-select

Level 3

Select **all** expressions equal to **35%** of **240**.

- **A.** 0.35 × 240
- **B.** (35/100) × 240
- **C.** 30% of 240 + 5% of 240
- **D.** 240 ÷ 0.35

Verify each selected expression

A, B, C: each equals 84.
D divides instead of multiplying.

☐ I evaluated every option.

C8 · Direction

Level 3

A value changes from **120** to **150**, then returns to **120**. Which change has the greater percent magnitude?

- **A.** The increase
- **B.** The decrease
- **C.** They are equal
- **D.** Not enough information

Show both original values

A: 30/120 = 25%; 30/150 = 20%.

☐ I used a new launchpad for each change.

C9 · Reverse percent

Level 4

72 is **30%** of what number?

Write 0.30 × whole = 72

0.30 × whole = 72
Whole = 72 ÷ 0.30 = 240
Check: 30% of 240 = 72.

Whole number **240**

☐ I checked 30% of my answer.

C10 · Error analysis

Level 4

To find the percent increase from **240** to **300**, Maya computes **60 ÷ 300**. Identify and correct her first wrong decision.

Name the wrong denominator and replace it

The first wrong decision is using 300, the new value, as the denominator.
Use the original value, 240:
60 ÷ 240 = 0.25 = 25%.

☐ My explanation names the original value.

C11 · Context

Level 3

At a community event, **45%** of **440** attendees arrived before noon. How many arrived at noon or later?

Find the requested group, not the given group

At noon or later: 100% − 45% = 55%.
55% of 440 = 220 + 22 = 242.

People **242**

☐ I answered the complement asked for.

C12 · Two changes

Level 4

A count moves from **200** to **230**, then to **207**. Compare the percent increase in the first change with the percent decrease in the second.

Calculate each percent from its own original

30/200 = 15% increase; 23/230 = 10% decrease.
The increase is larger by 5 percentage points.

Comparison sentence **15% increase > 10% decrease**

☐ I used 200 first and 230 second.

D SHSAT Transfer Set

8 questions · 20 min

Model the full situation before choosing an operation. The final line should contain only the requested quantity.

D1 · Part of a part Level 3

A museum sold **680** tickets. Of the tickets, **35%** were sold online. One fourth of the online tickets were student tickets. How many online student tickets were sold?

Whole → online part → student part

Online tickets: 0.35 × 680 = 238.
Online student tickets: 238 ÷ 4 = 59.5.

The exact data are inconsistent: tickets must be counted in whole numbers. No whole-ticket count satisfies the original item. Do not round to 60.

Tickets **59.5 (inconsistent ticket count)**

D2 · Percent then fraction Level 3

In a survey of **250** students, **72%** chose an after-school activity. Of those students, **5/9** chose a sport. How many surveyed students chose a sport?

Find the subgroup in two stages

Activity subgroup: 0.72 × 250 = 180.
Sport subgroup: (180 ÷ 9) × 5 = 100.

The fraction 5/9 applies to the 180 activity students, not all 250.

Students **100**

D3 · Geometry transfer Level 3

A rectangle’s width stays fixed while its length increases from **25** units to **31** units. By what percent does the rectangle’s area increase?

Use area structure; a width value is unnecessary

Let the fixed width be w.
Old area = 25w; new area = 31w.
Change = 6w.
6w/(25w) = 6/25 = 24%.

With width fixed, the area increases by the same percent as the length.

Percent number only **24**

D4 · Ratio transfer Level 4

The angles of a triangle are in the ratio **3:5:7**. The smallest angle is what percent of the largest angle?

The common scale factor may cancel

Smallest angle = 3k; largest = 7k.
3k/(7k) = 3/7.
Percent = (3/7) × 100% = (300/7)%.

Exact percent number: 300/7
Decimal: 42.857142…
No rounding precision is specified.

Percent number only **300/7**

D SHSAT Transfer Set (continued)

8 questions · 20 min

Model the full situation before choosing an operation. The final line should contain only the requested quantity.

D5 · Statistics transfer

Level 3

The mean of five quiz scores is **64**. A sixth score of **88** is added. By what percent does the mean increase?

Recover the old total, then find the new mean

Old total = 5 × 64 = 320.
New total = 320 + 88 = 408.
New mean = 408 ÷ 6 = 68.
Increase in mean = 68 − 64 = 4.
4/64 = 1/16 = 0.0625 = 6.25%.

Percent number only 6.25

D6 · Multi-select

Level 3

Select **all** values that are greater than **37%** but less than **0.4**.

A. 3/8
B. 0.385
C. 39.5%
D. 2/5

Convert to one representation

Answer: A, B, C
3/8 = 0.375
0.385 = 0.385
39.5% = 0.395
All are between 0.37 and 0.40.
2/5 = 0.40, so D is excluded.

D7 · Two categories

Level 4

A store has **360** pens. Blue pens are **40%** of the total. Black pens are **25%** of the pens that are not blue. How many pens are neither blue nor black?

Track the changing whole carefully

Not blue: 60% of 360 = 216.
Black: 25% of 216 = 54.
Neither blue nor black: 216 − 54 = 162.

Check: 144 + 54 + 162 = 360.
The 25% uses 216 as its whole.

Pens 162

D8 · Claim analysis

Level 4

A student says, “If a population increases **20%** one year and decreases **20%** the next year, it returns to its starting value.” Test the claim using a convenient starting value and explain the denominator shift.

Example + conclusion + two original values

False. Start with 100.
Increase 20%: 100 + 20 = 120.
Decrease 20%: 120 − 24 = 96.
Net change: a 4% decrease.

The first 20% uses 100 as its base; the second uses 120. Equal rates remove different amounts because the bases differ.

E One-Screen Rehearsal

3 decisions · 5 min

Imagine the Next button locks the current item. State the final check you would perform in each situation.

E1. Your answer for **18%** of a positive number is greater than the original number. What structural check should interrupt your commit?

State the expected size relationship.

Stop and recompute: 18% of a positive whole must be less than that whole ($0 < 0.18N < N$).

E2. Your percent decrease is negative, but the prompt asks for the magnitude of the change. What belongs in the response field?

Separate direction from magnitude.

Enter the nonnegative percent magnitude; omit the percent sign if a number-only entry is required.

E3. The prompt asks how many students, but your field contains “34%.” Identify the mismatch and state your next action before advancing.

Name the requested quantity and correct the form.

A count is needed, not a rate. Replace 34% with 0.34 × the correct whole; the whole is needed for a count.

F Process Debrief

Evidence, not impressions

Complete this before checking solutions. Focus on decisions you can repeat under forward-only conditions.

- ☐ I named the requested quantity before calculating.
  C11 asks for the 55% complement.
- ☐ I found the change before dividing on percent-change items.
  C5 starts with 560 − 434 = 126.
- ☐ I predicted whether the answer should be smaller or larger.
  C2 must be below half of 160 (= 80).
- ☐ I entered only the value and form requested.
  A3 converts 0.285 to the entry 28.5.
- ☐ I identified the whole for every percent-of problem.
  D2 uses 250, then the subgroup of 180.
- ☐ I divided each change by its own original value.
  C12 uses 200 first, then 230.
- ☐ I used benchmarks to audit exact arithmetic.
  C1 uses 87.5 − 3.5 = 84.
- ☐ I committed after one evidence-based final check.
  C9 checks that 30% of 240 is 72.

Timing record

Record your time by section and identify where your pace changed.

A–B _____ **C** _____

D–E _____ **Total** _____

Record actual times; no fixed answer.

Confidence audit

List two items you would most want to revisit on a non-adaptive test. Then explain the verification you used instead.

Item numbers + substitute verification

C9: 30% of 240 = 72.
D5: 408/6 = 68; (68 − 64)/64 = 6.25%.

G Error Log

Up to 5 corrections

Use one primary type: Concept, Procedure, Reading, Arithmetic, Format, or Wrong Quantity.

#	Problem	First wrong decision	Primary type	Next-time interruption cue
1	C10 (sample)	I used 300, the new value, as the denominator.	Concept	Circle the “from” value before dividing.
2	C6 (sample)	I subtracted 27 from 180 although “by 27” gives the change.	Reading	Label “by 27” as change = 27.
3	C5 (sample)	I calculated 560 − 434 as 136.	Arithmetic	Add back: 434 + 126 = 560.
4	A3 (sample)	I left 0.285 in a percent-number field.	Format	Read “percent number”; multiply by 100 once.
5	C11 (sample)	I stopped at 198, the before-noon group.	Wrong Quantity	Underline “at noon or later”; find the complement.

H Transfer Reflection

Complete sentences

1. The benchmark that made one exact calculation easier was…

25% − 1% on C1. One quarter of 350 is 87.5; subtracting 3.5 gives 84 without long multiplication.

2. The words that tell me which value is the original are…

“From,” “original,” and “before the change.” In C12, “then to 207” starts a new change whose original value is 230.

3. One typed-entry mismatch I can now catch is…

A decimal proportion is not the percent number: 0.285 must become 28.5 when the field asks for a percent number only.

4. My five-second percent-change commit question will be…

“Did I divide the actual change by the value before this change, convert to percent units, and enter the requested magnitude?”